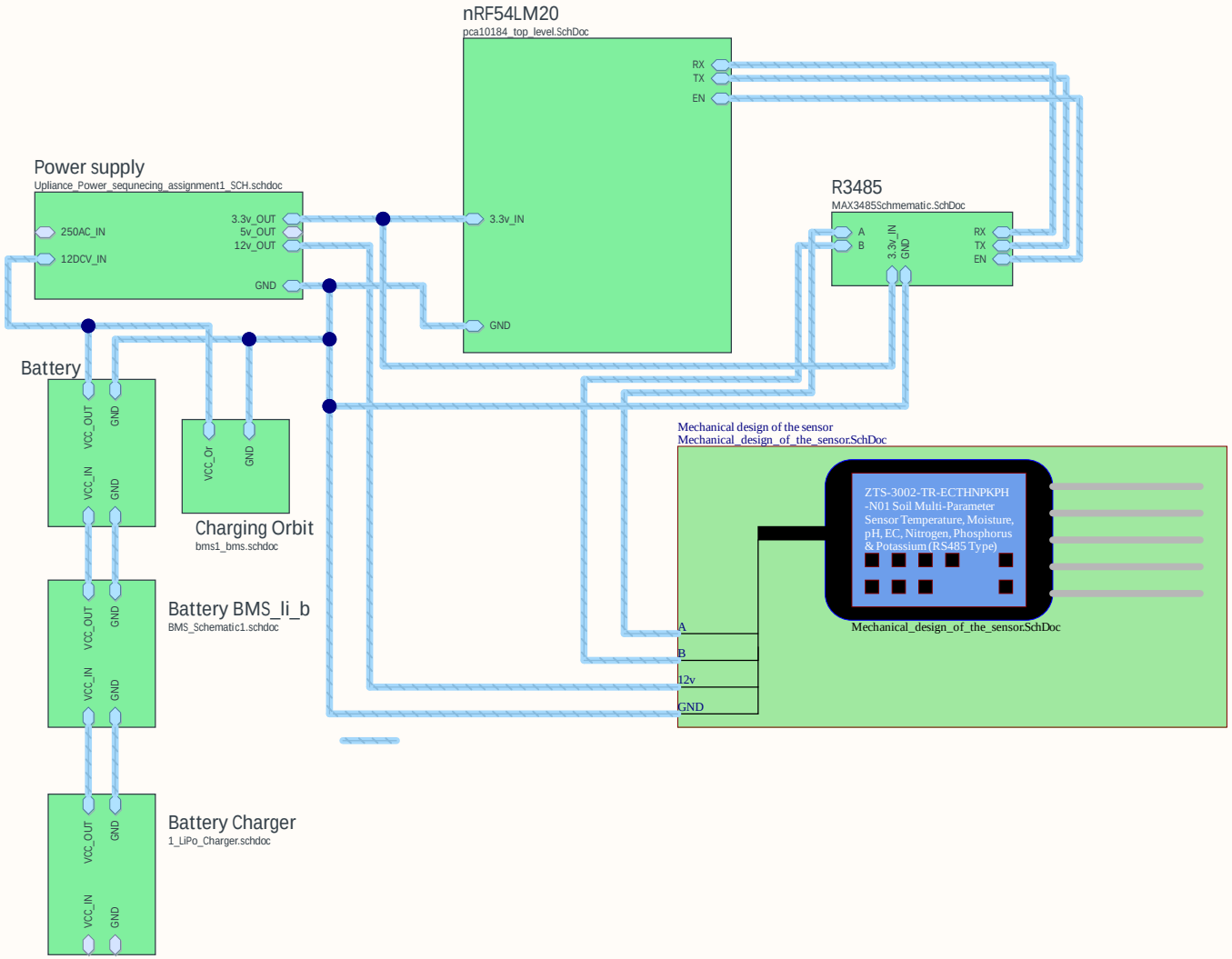
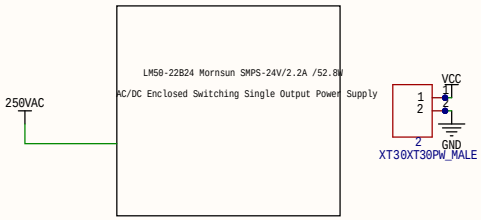

nRF54LM20
pca10184_top_level.SchDoc
RX
TX
EN
3.3v_IN
GND
Power supply
Upliance_Power_sequnecing_assignment1_SCH.schdoc
250AC_IN
12DCV_IN
3.3v_OUT
5v_OUT
12v_OUT
GND
R3485
MAX3485Schmematic.SchDoc
A
B
3.3v_IN
GND
RX
TX
EN
Battery
VCC_OUT
GND
VCC_IN
GND
VCC_Or
GND
Charging Orbit
bms1_bms.schdoc
Mechanical design of the sensor
Mechanical_design_of_the_sensor.SchDoc
ZTS-3002-TR-ECTHNPKPH
-N01 Soil Multi-Parameter
Sensor Temperature, Moisture,
pH, EC, Nitrogen, Phosphorus
& Potassium (RS485 Type)
Mechanical_design_of_the_sensor.SchDoc
A
B
12v
GND
Battery BMS_li_b
BMS_Schematic1.schdoc
Battery Charger
1_LiPo_Charger.schdoc

POWER SEQUENCE DESIGN

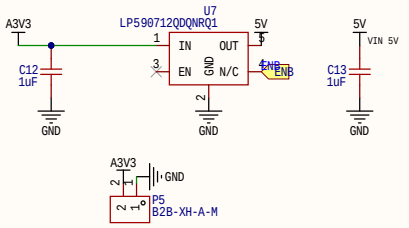
AC TO DC POWER SEQUENCYING

250 VAC TO 24V DC

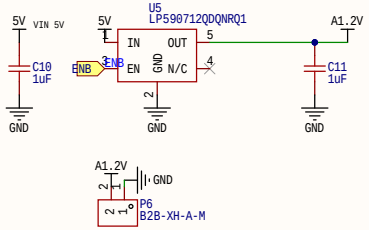


DC TO DC ANALOG POWER SEQUENCING

5V TO 3.3 ANALOG

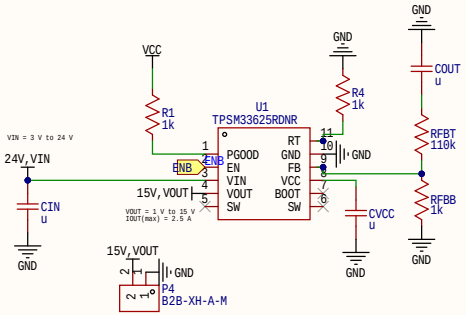


5V TO 1.2V ANALOG

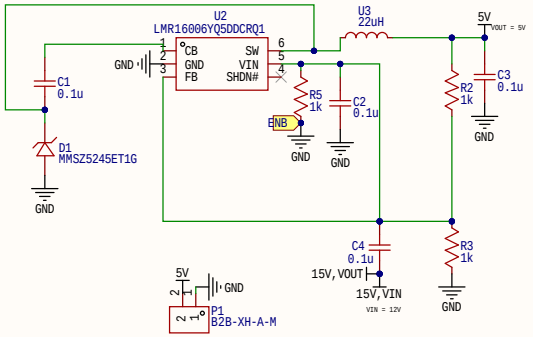


DC TO DC POWER SEQUENCING

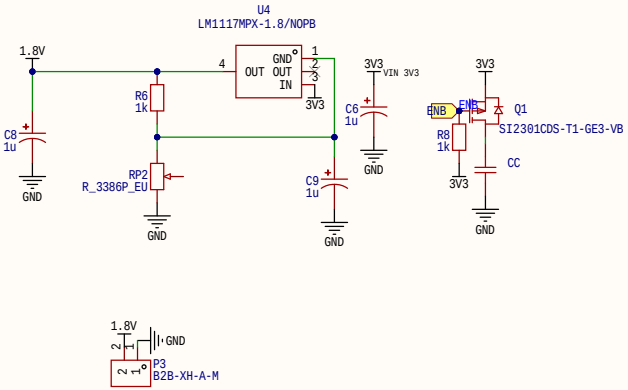
24V TO 12V 2.5A



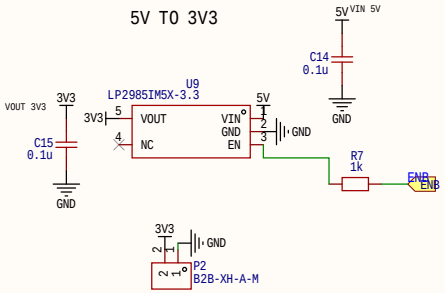
12V TO 5V



3V3 TO 1.8V

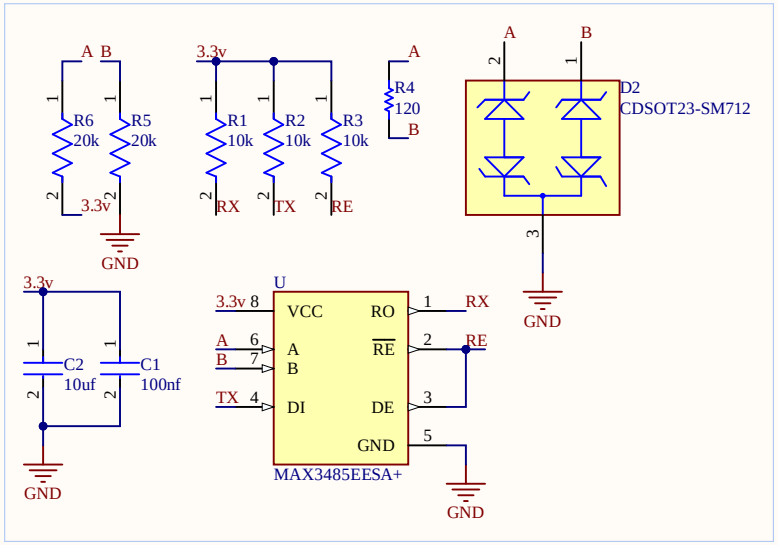


5V TO 3V3



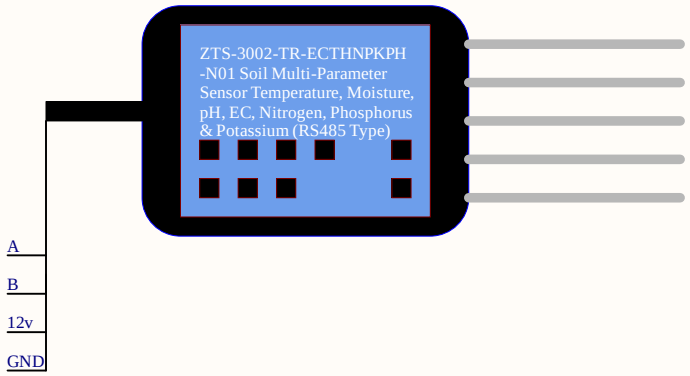
TITLE: POWER SEQUENCING		REV: 1.0
	Company:	Sheet: 1/1
	Date: 2025-04-19	Drawn By: janardhan bv

RS3485-485



Title		
Size A4	Number	Revision
Date: 10/18/2026		Sheet of
File: C:\Users\..\MAX3485Schmematic.SchDoc		Drawn By: Yusuf Yasser

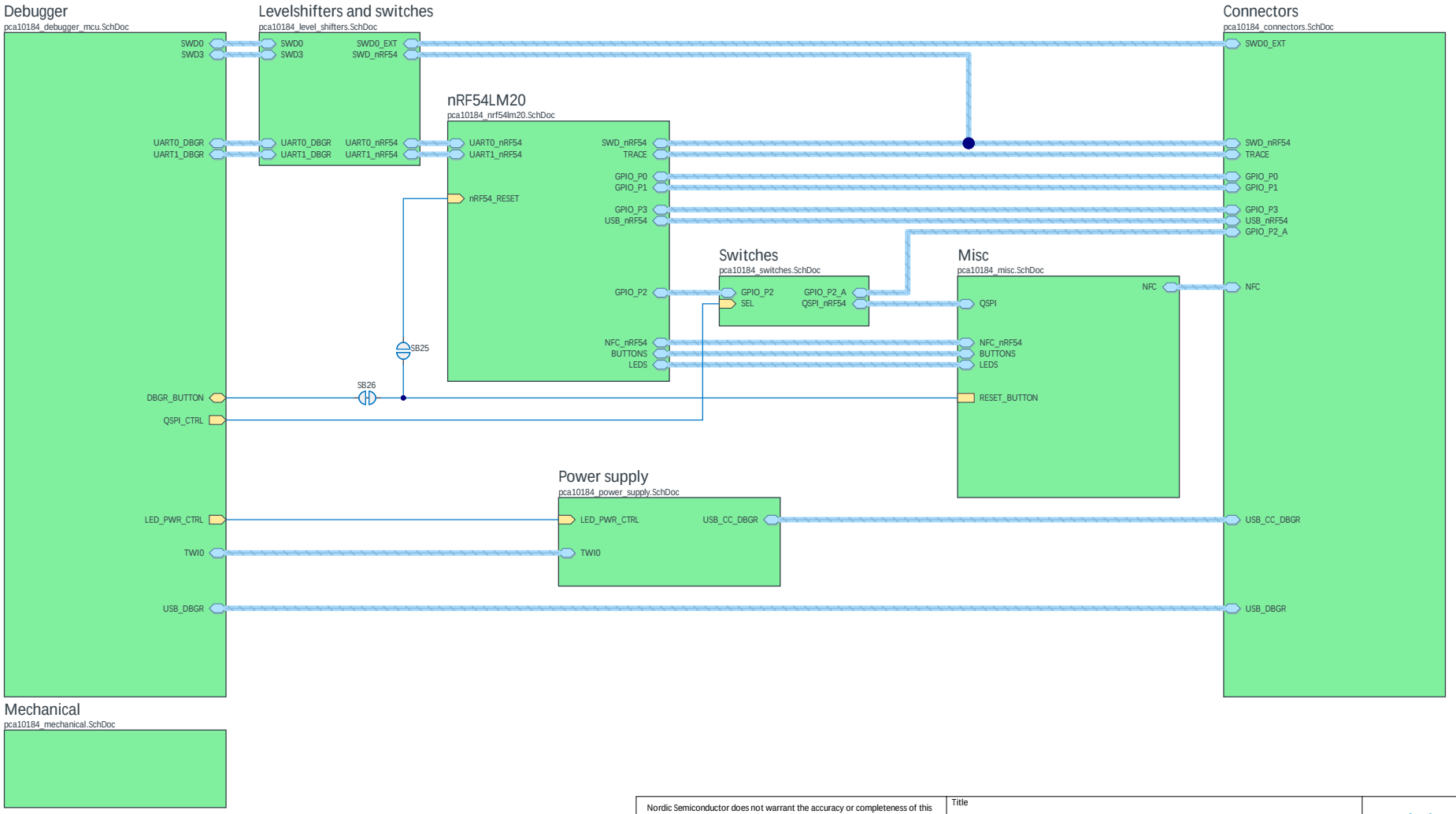
Soil Multi-Parameter Sensor



Mechanical diagram illustrating the sensor's shape

Nordic Semiconductor ASA

()

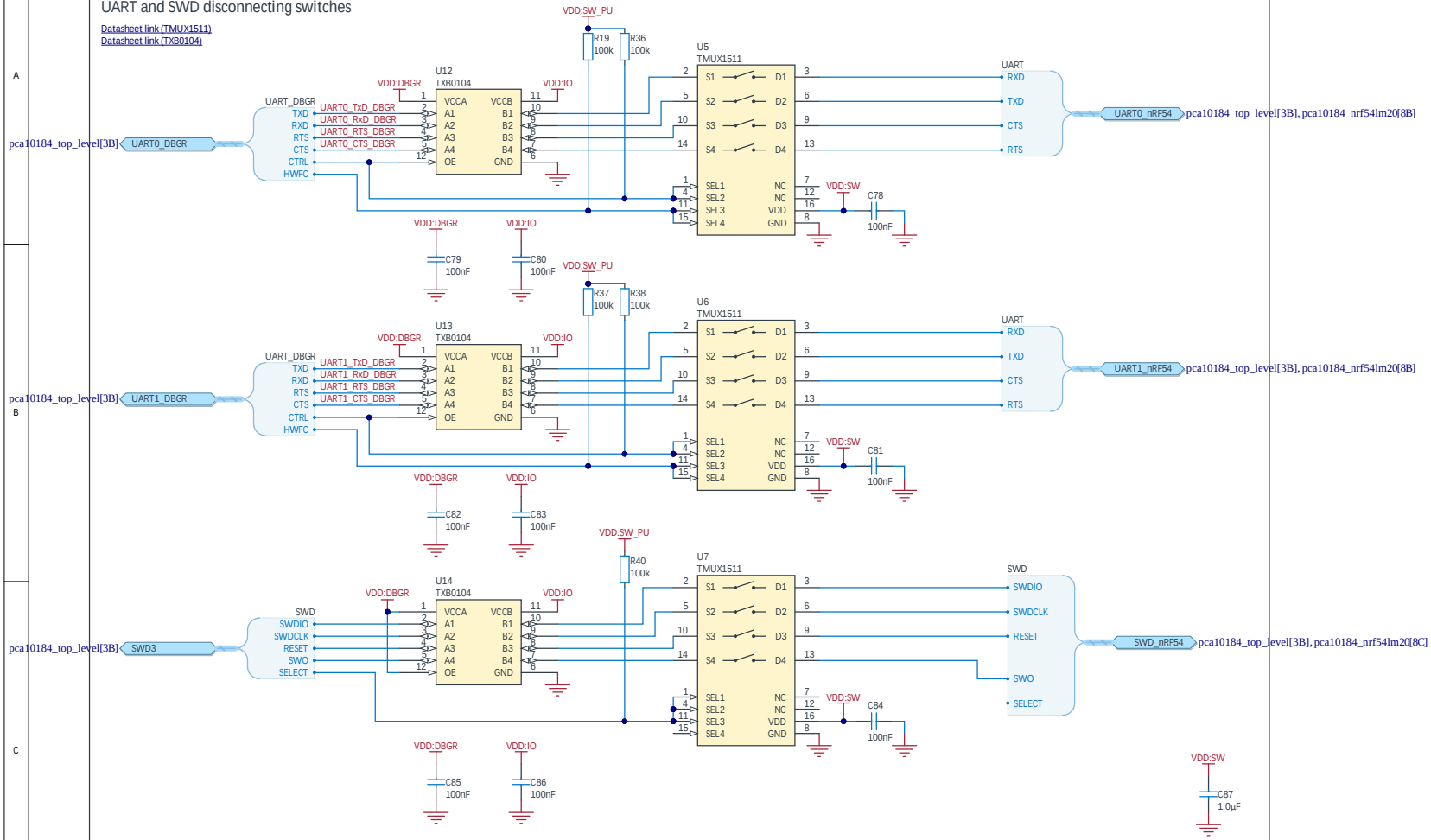


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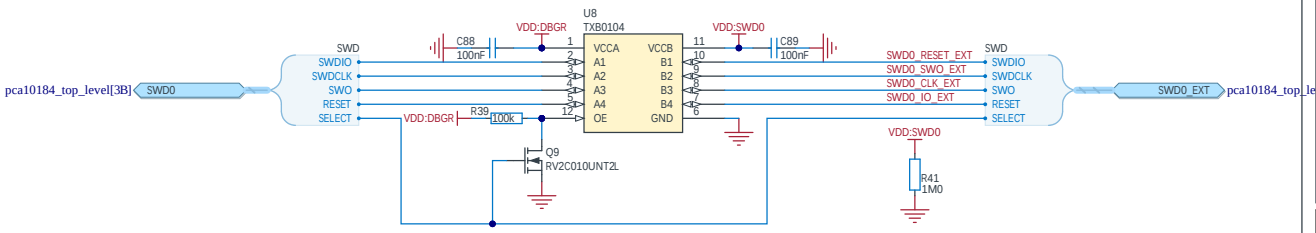
Title		
Size: A3	PCB Assembly Number	Revision
Date: 08/10/2026		Sheet 5 of 17
File: pca10184_top_level.SchDoc		Drawn By:
Classification:		

UART and SWD disconnecting switches

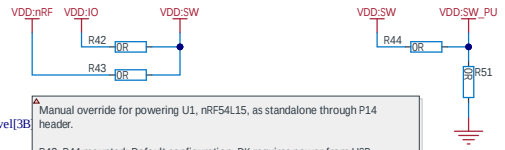
Datasheet link (TMUX1511)
Datasheet link (TXB0104)



SWD Debug Out level shifter



Analog switches manual override



Manual override for powering U1, nRF54L15, as standalone through P14 header.

R42, R44 mounted: Default configuration. DK requires power from USB connector.
R43, R51 mounted: Analog switches are powered from P14. Analog switches U5, U6, U7, U15 and U16 disconnects all signals. No external power needed from USB connector

Title: - Level Shifters		
Size: A3	PCB Assembly Number:	Revision:
Date: 08/10/2026		Sheet 6 of 17
File: pca10184_level_shifters.SchDoc		Drawn By:
Classification:		

NORDIC SEMICONDUCTOR

The No ERC object is a design directive.
This directive is placed on a node in the circuit to suppress harmless warnings and/or erro r violation conditions that are detected when the schematic project is compiled.

Board fiducials

Logos and markings

<year>.<week>
<UID>



WEE Symbol

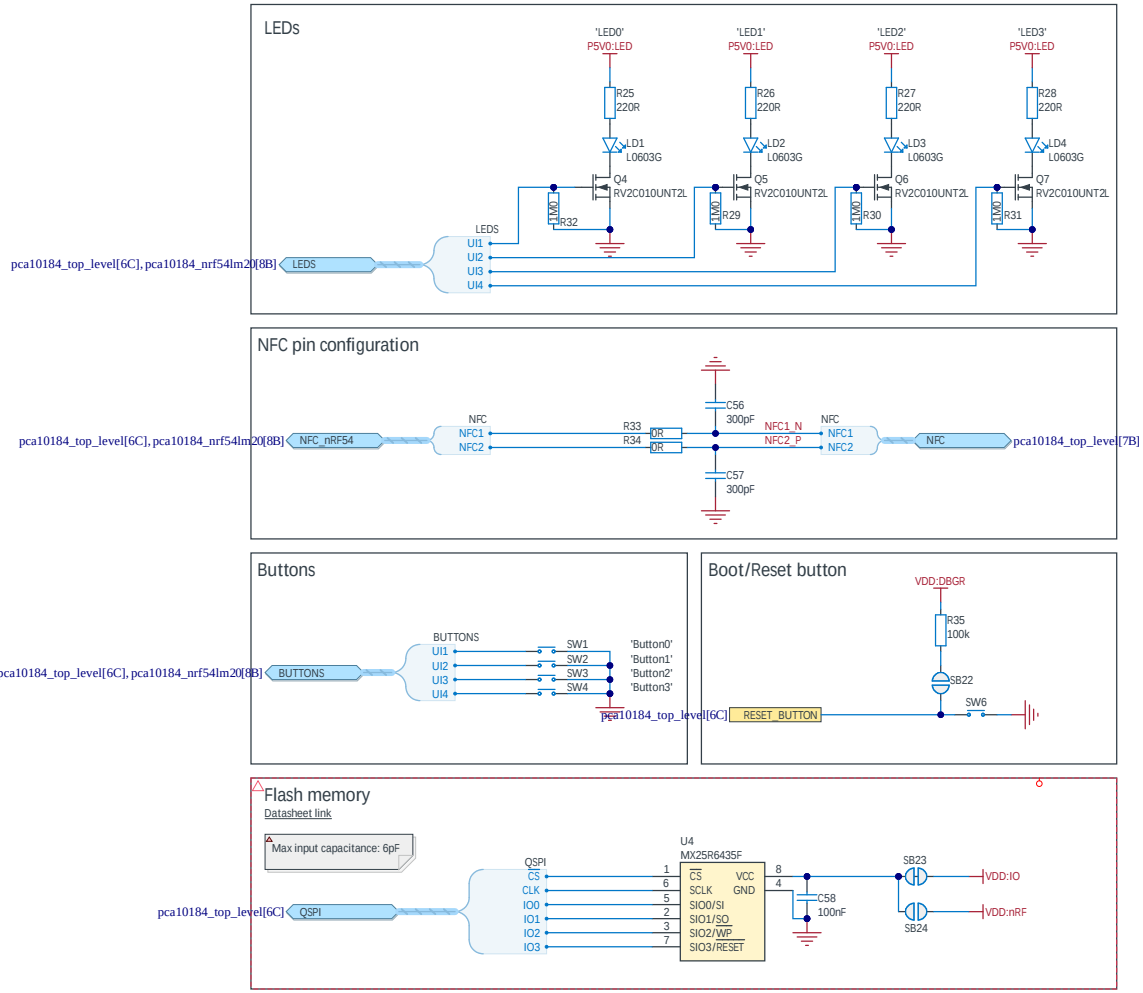


Mechanical holes



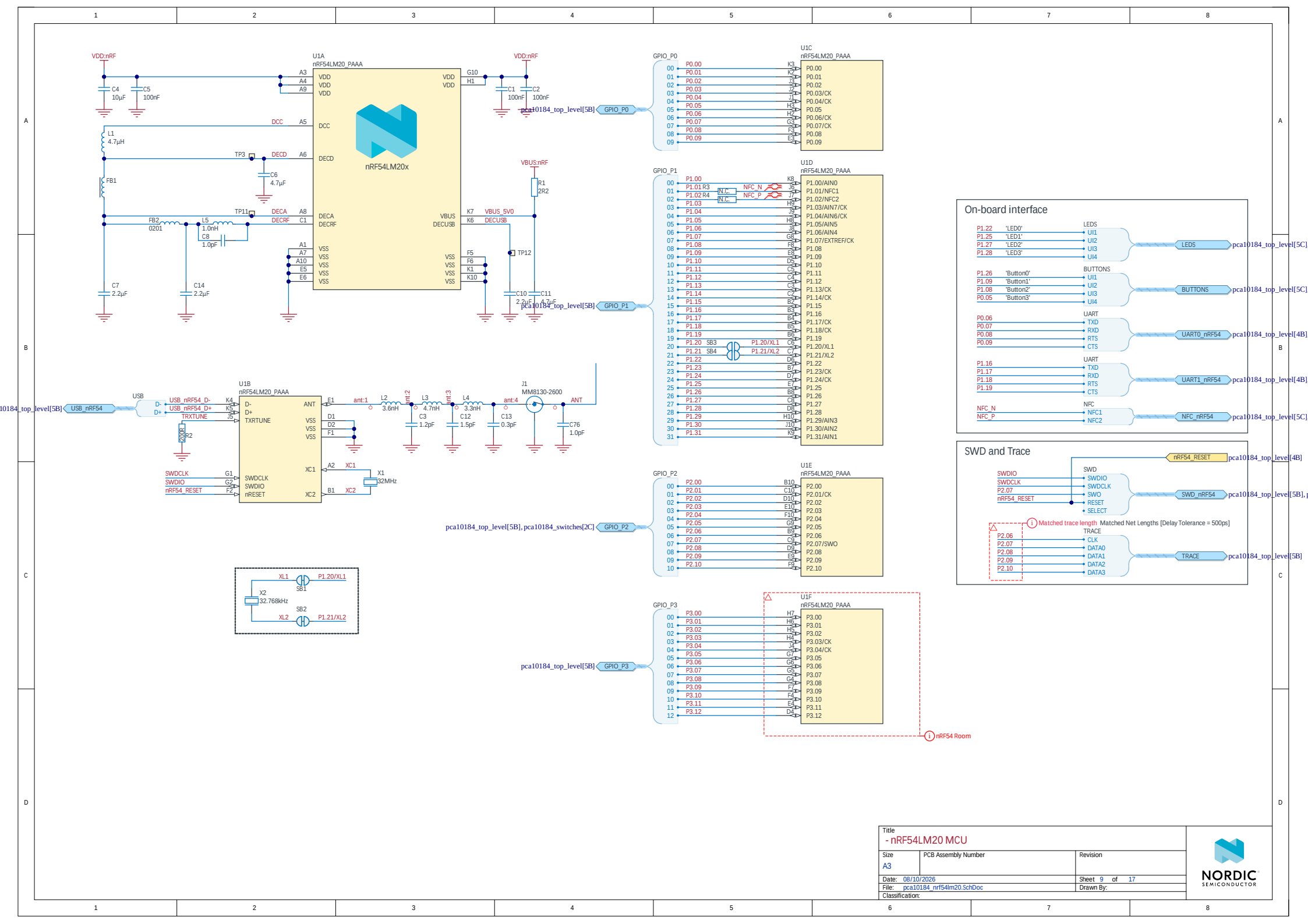
Title - Mechanical		
Size A4	PCB Assembly Number	Revision
Date: 08/10/2026		Sheet 7 of 17
File: pca10184_mechanical.SchDoc		Drawn By:
Classification:		

NORDIC SEMICONDUCTOR



Title	- Misc	
Size	PCB Assembly Number	Revision
A3		
Date: 08/10/2026		Sheet 8 of 17
File: pca10184_misc.SchDoc		Drawn By:
Classification:		

NORDIC SEMICONDUCTOR


U1A
nRF54LM20_PAAA
nRF54LM20x
VDD:nRF
VBUS:nRF
DCC
DECD
DECA
DECRF
VBUS
DECUSB
VSS
C4 10µF
C5 100nF
C1 100nF
C2 100nF
L1 4.7µH
FB1
TP3
C6 4.7µF
TP11
FB2 0201
L5 1.0nH
C8 1.0pF
C7 2.2µF
C14 2.2µF
R1 2R2
TP12
C10 2.2µF
C11 4.7µF
U1B
USB
USB_nRF54_D-
USB_nRF54_D+
TRXTUNE
R2 200R
SWDCLK
SWDIO
nRF54_RESET
ANT
XC1
XC2
X1 32MHz
L2 3.6nH
L3 4.7nH
L4 3.3nH
C3 1.2pF
C12 1.5pF
C13 0.3pF
C76 1.0pF
J1 MM8130-2600
ant:1
ant:2
ant:3
ant:4
XL1
XL2
SB1
SB2
X2 32.768kHz
P1.20/XL1
P1.21/XL2
U1C
U1D
U1E
U1F
GPIO_P0
GPIO_P1
GPIO_P2
GPIO_P3
pca10184_top_level[5B]
pca10184_top_level[5B], pca10184_switches[2C]
NFC_N
NFC_P
On-board interface
LEDS
'LED0'
'LED1'
'LED2'
'LED3'
BUTTONS
'Button0'
'Button1'
'Button2'
'Button3'
UART
TXD
RXD
RTS
CTS
UART0_nRF54
UART1_nRF54
NFC
NFC1
NFC2
NFC_nRF54
SWD and Trace
SWD
SWDIO
SWDCLK
SWO
RESET
SELECT
SWD_nRF54
TRACE
CLK
DATA0
DATA1
DATA2
DATA3
Matched trace length
Matched Net Lengths [Delay Tolerance = 500ps]
nRF54 Room
Title
- nRF54LM20 MCU
Size
A3
PCB Assembly Number
Revision
Date: 08/10/2026
Sheet 9 of 17
File: pca10184_nrf54lm20.SchDoc
Drawn By:
Classification:
NORDIC SEMICONDUCTOR

Power managment

U3
nPM1300-QEAA

nPM1300

VBUS
R14 0R
C34 4.7µF
C35 100nF
C36 4.7µF
C37 100nF
USB_CC
USB_CC_DBGR
CC1
CC2
VBAT
NTC
TP35 LDO1_CTRL
LED_PWR_CTRL
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
Startup VOUT1: 47k -> 1.8V
Startup VOUT2: 4.7k -> 1.8V
R17 47k
R18 4k7
VSET1
VSET2
SHPHLD
TWI
TWI0
SCL
SDA
TWI0_SCL
TWI0_SDA
VDD:DBGR
C54 100nF
VDDIO
VBUSOUT
VSYS
PVDD
C39 1.0µF
C40 10µF
C41 10µF
C42 1.0nF
C43 1.0nF
PVSS1
PVSS2
C44 10µF
SB18
C45 10µF
SB17
R16 0R
C38 4.7µF
TP4
P5V0
VDD:nRF_PMIC
VOUT1
L7 2.2µH
C46 10µF
SW1
TP5
C47 100nF
SB20
VDD:IO
VDD:nRF = VDD:IO = 1.8 - 3.6V
Via to GND-layer on PVSS1
VOUT2
L6 2.2µH
C48 10µF
SW2
TP6
C49 100nF
SB21
Via to GND-layer on PVSS2
LSIN1/VINLDO1
LSIN2/VINLDO2
SB27
SB28
SB29
SB30
LSOUT1/VOUTLDO1
LSOUT2/VOUTLDO2
LDO1
TP1
P5V0:LED
C33 10µF
C50 10µF
C51 10µF
C52 10µF
AVSS

Current Measurement

VDD:nRF_PMIC
VDD:nRF
VDD:IO
R15 10R
P14
Pin List 1x3
JP1
Shunt 2.54mm
SB19

External power source

P16
Pin List 1x3
VBUS
5V supply to the board, in case USB is not used. Using the solder bridge VBUS-5V0 it's possible to bypass PMIC as well.

nPM1300 settings:
- VOUT1: 1.8V - 3.3V, max 200mA
- VOUT2: 1.8V - 3.3V, max 200mA
- LS2: Switch for LEDs power. GPIO1 control

VSYS current limit: 900mA

Power Switch

VBUS
VBUS:DBGR
Q2A MCM2301-TP
Q2B MCM2301-TP
TP14
R20 1M0
R21 10k
C55 4.7µF
SW5
Switch
R22 10k
R23 100k
VBUS:nRF
SB6
R24 1M0
Q3A MCM2301-TP
Q3B MCM2301-TP

VDD:nRF supply buffer

C59 100nF
P5V0
Q8 PBSS4330PA
TP2
VDD:IO
R52 0R
R53 10k
VDD:nRF
33k R55
C60 100nF
U9 TLV271
R54
The VDD:nRF buffer is not needed for nRF54LM20 reference design.
Only used for DK operation and peripherals

VBUS
5V0:CONN
SB31

GND probe point

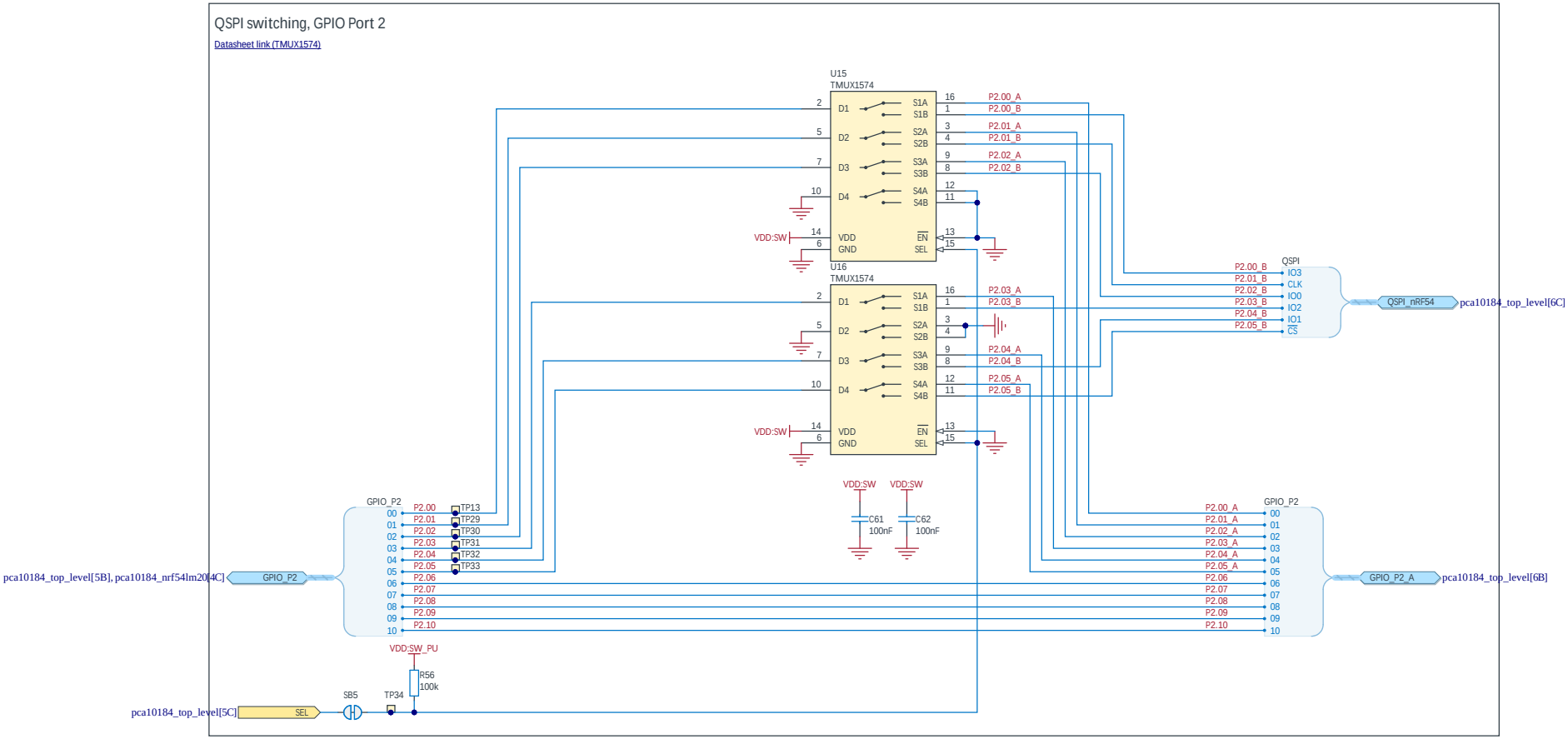
TP15

GND Test Points

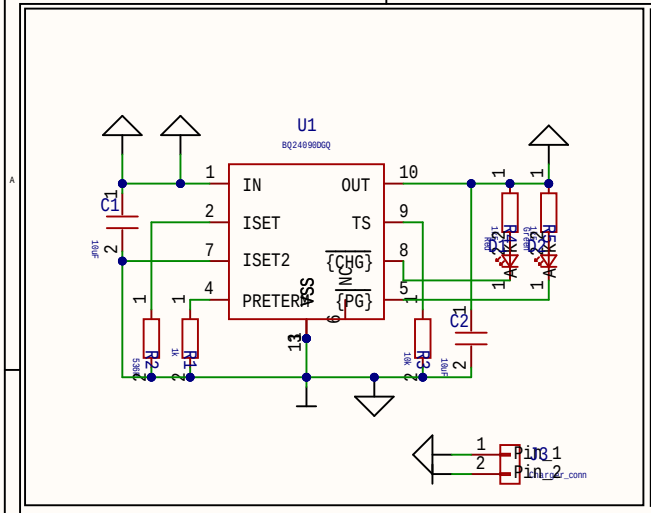
TP16 TP17 TP18 TP19 TP20

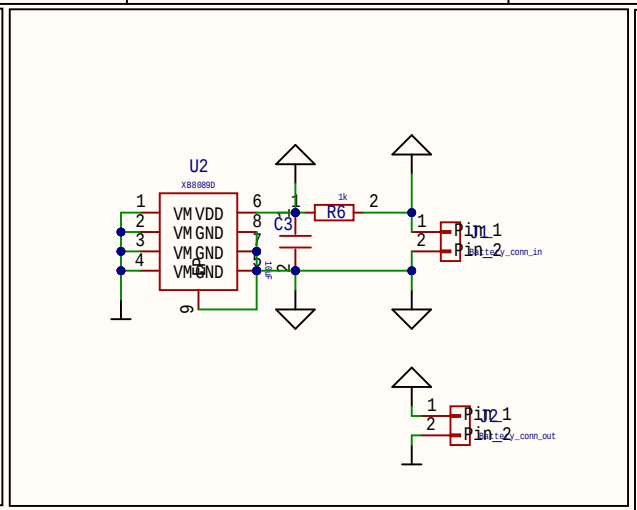
Title: - Power Supply		
Size: A3	PCB Assembly Number	Revision
Date: 08/10/2026		Sheet 10 of 17
File: pca10184_power_supply.SchDoc		Drawn By:
Classification:		

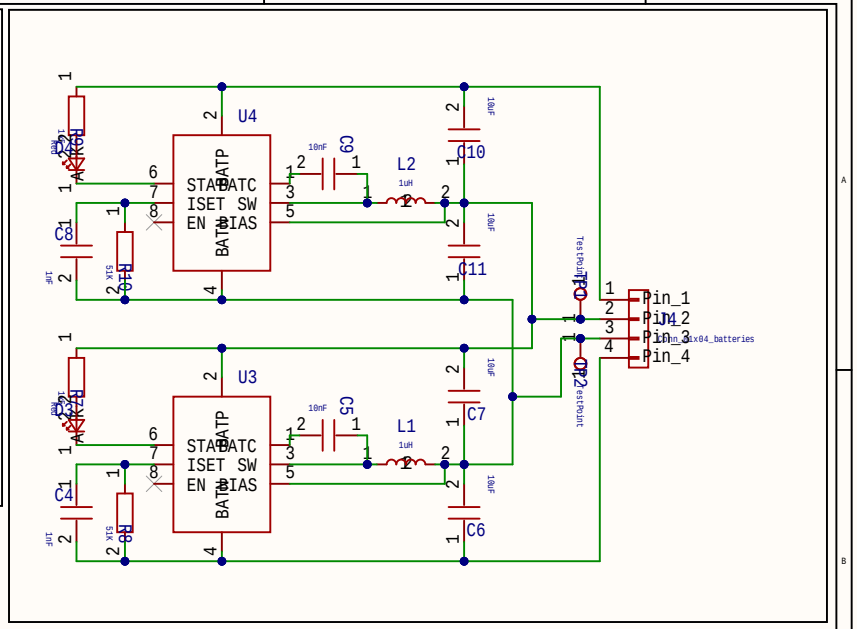
NORDIC SEMICONDUCTOR



Title	- QSPI Switching	
Size: A3	PCB Assembly Number	Revision
Date: 08/10/2026		Sheet 11 of 17
File: pca10184_switches.SchDoc		Drawn By:
Classification:		



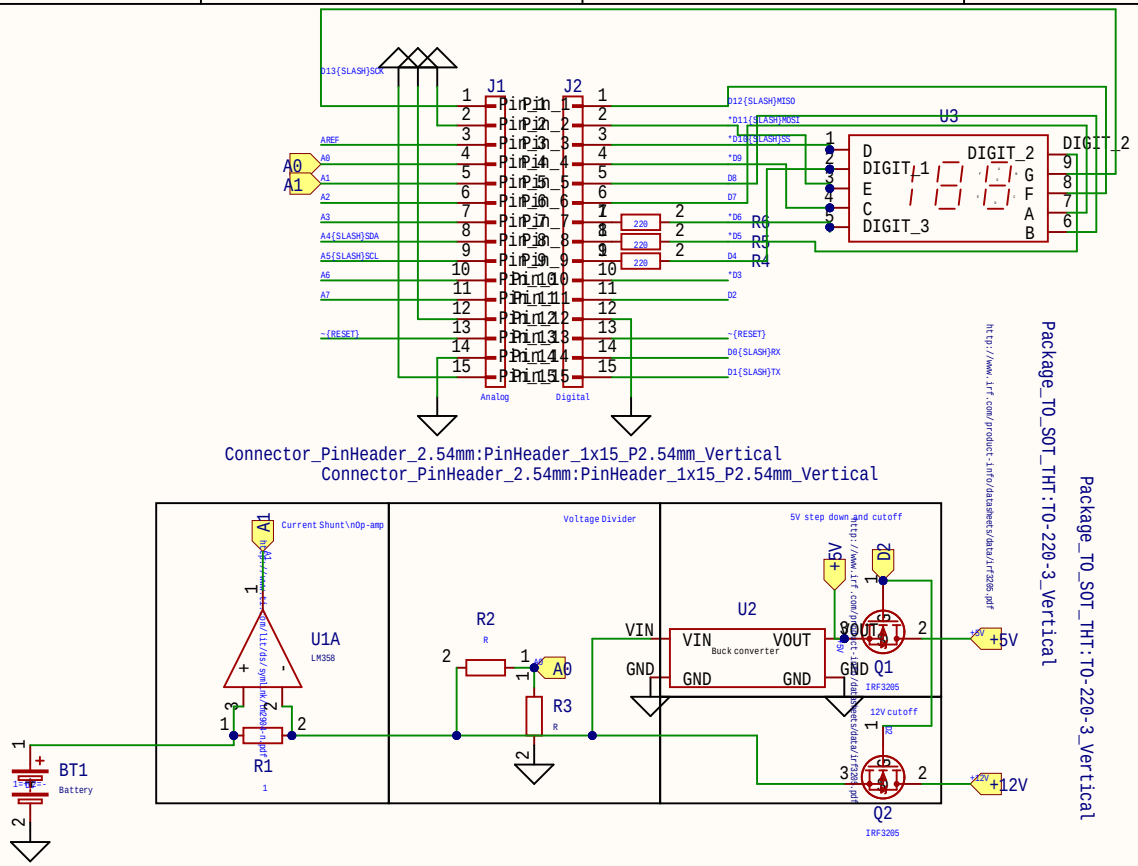




Battery Cahrger

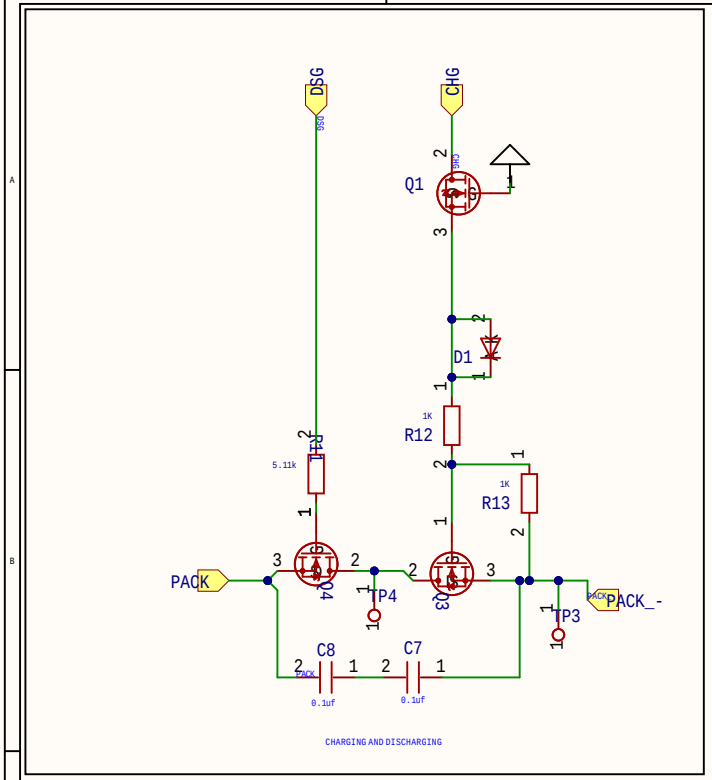
Sheet:/ File:LiPo Charger		
Title:		
Size:"A4"	Date:	Rev: Id:/

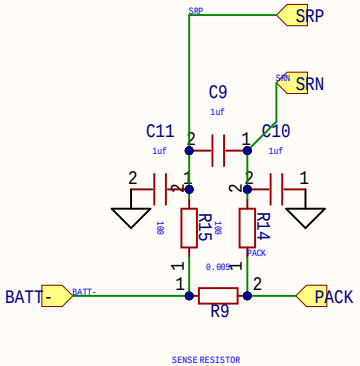
POWE

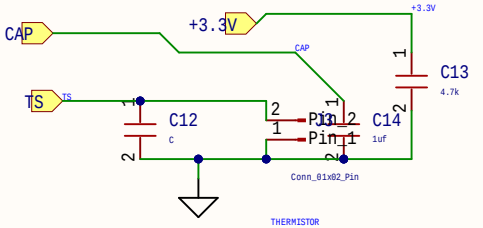


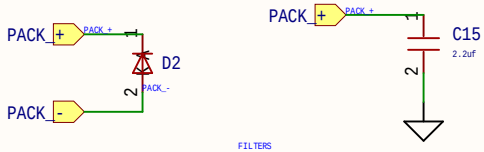
Charging Orbit

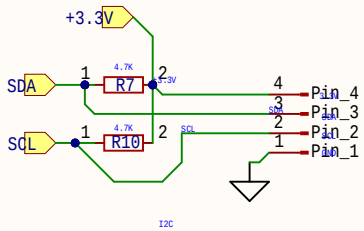
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File:bms		
Title:		
Size:"A4"	Date:	Rev:
		Id:/

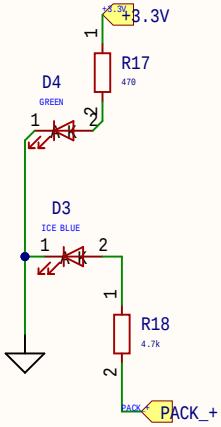


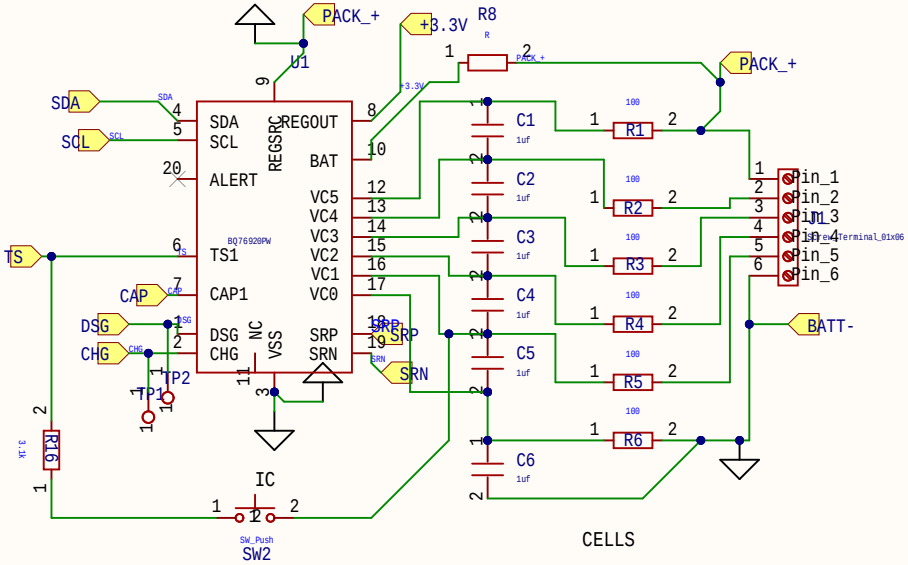










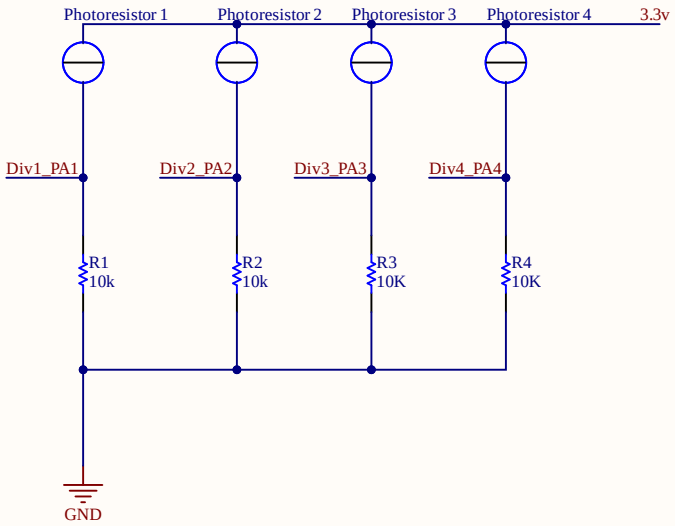


BMS Battery

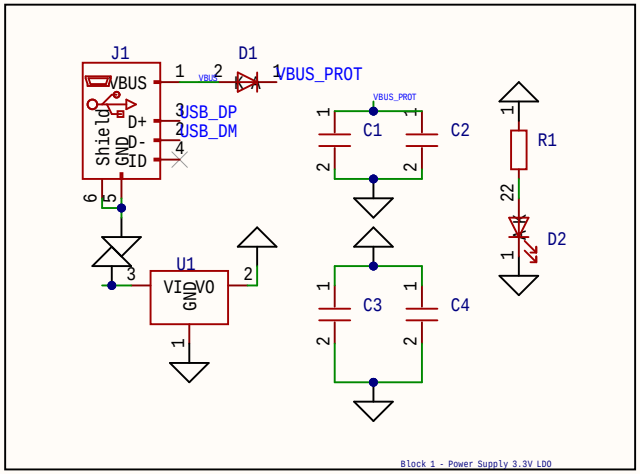
Sheet:/		
File:BMS Schematic		
Title:		
Size:"A4"	Date:	Rev:
		Id:/

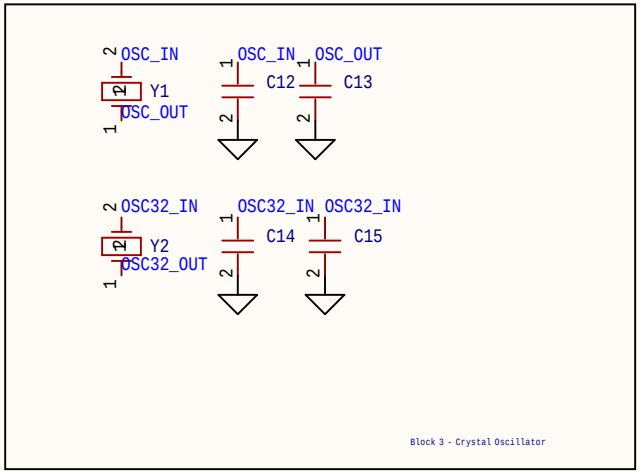
BATT

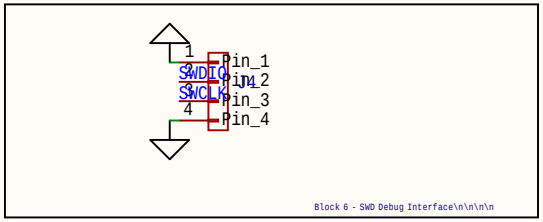
The readings from the three axes are processed by an STM32 microcontroller, which then controls the rotation direction of the solar tracker using a servo motor.

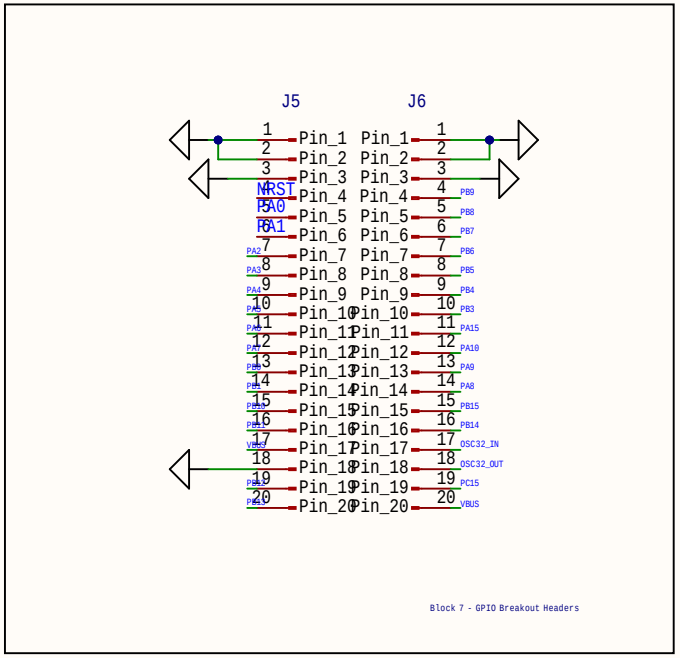


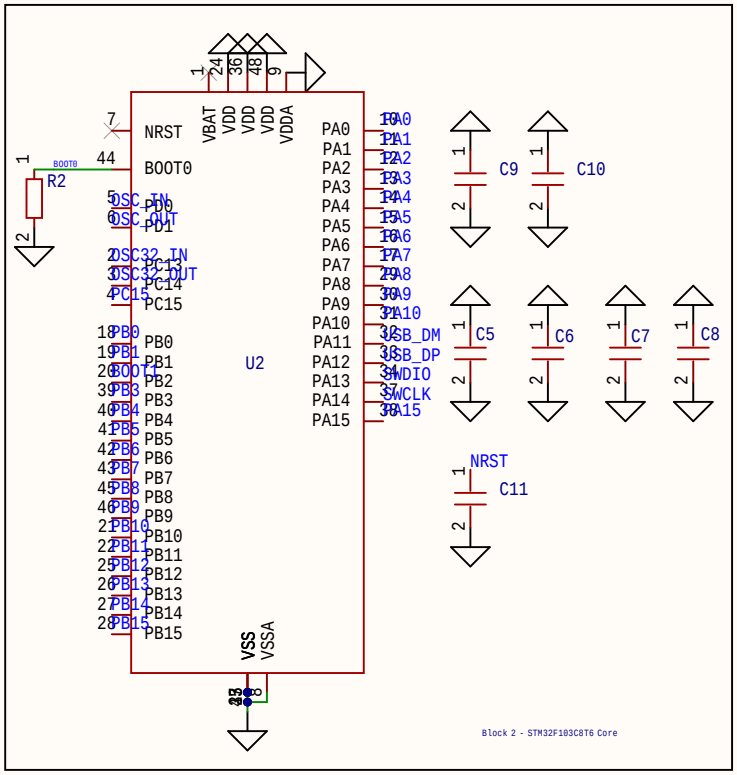
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Size: A4	Number	Revision
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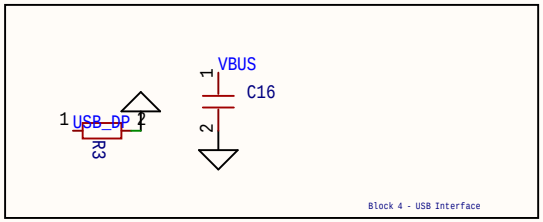


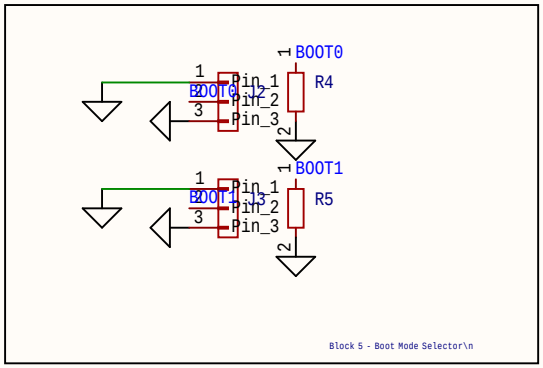












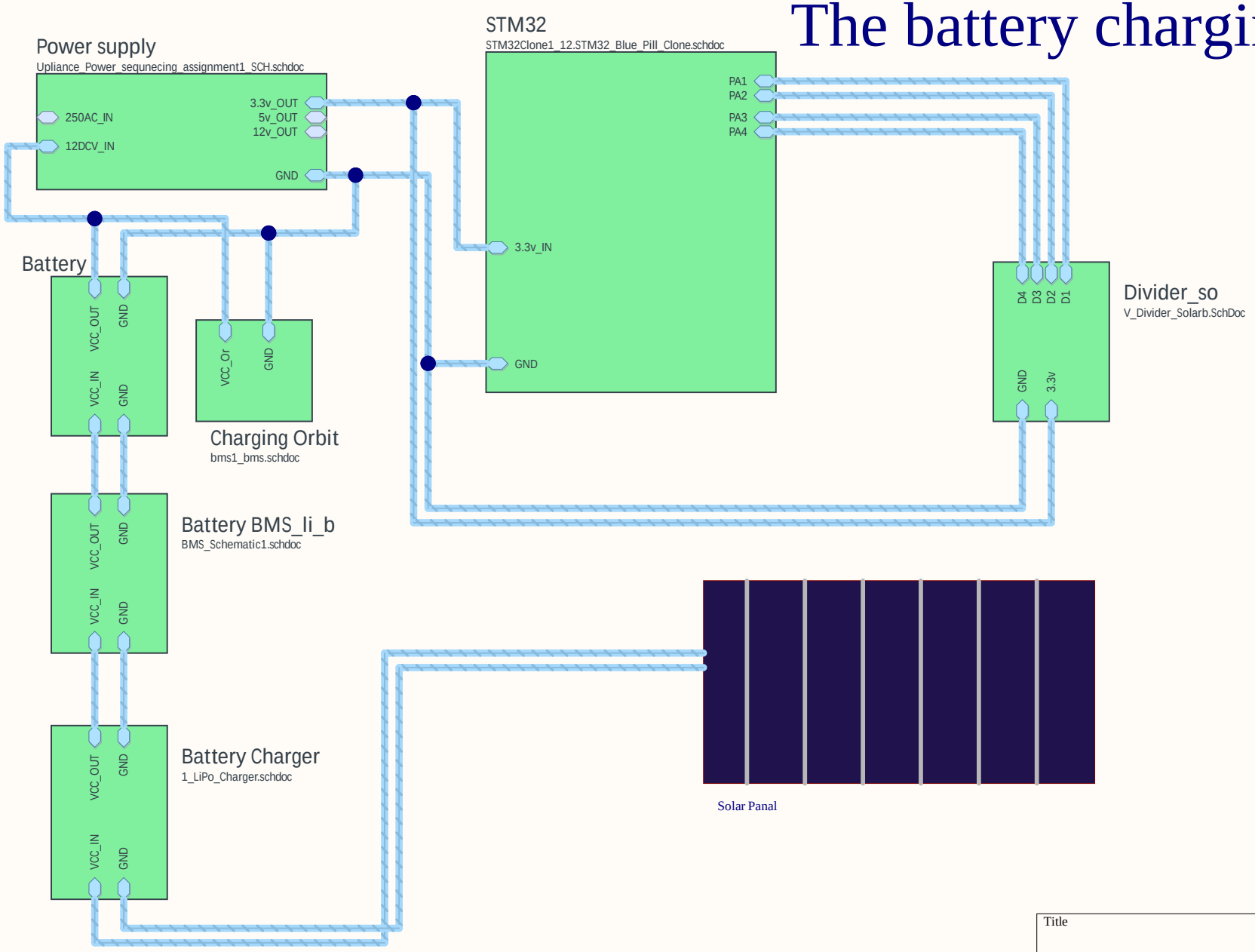
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The battery charging section



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Board Stack Report